

SOFTWARE ENGINEERING LAB-o1 (UE24CS341A)

TEAM NO – 001

PROBLEM STATEMENT – 001

Smart Lab Equipment Slot Reservation Portal

NAMES :

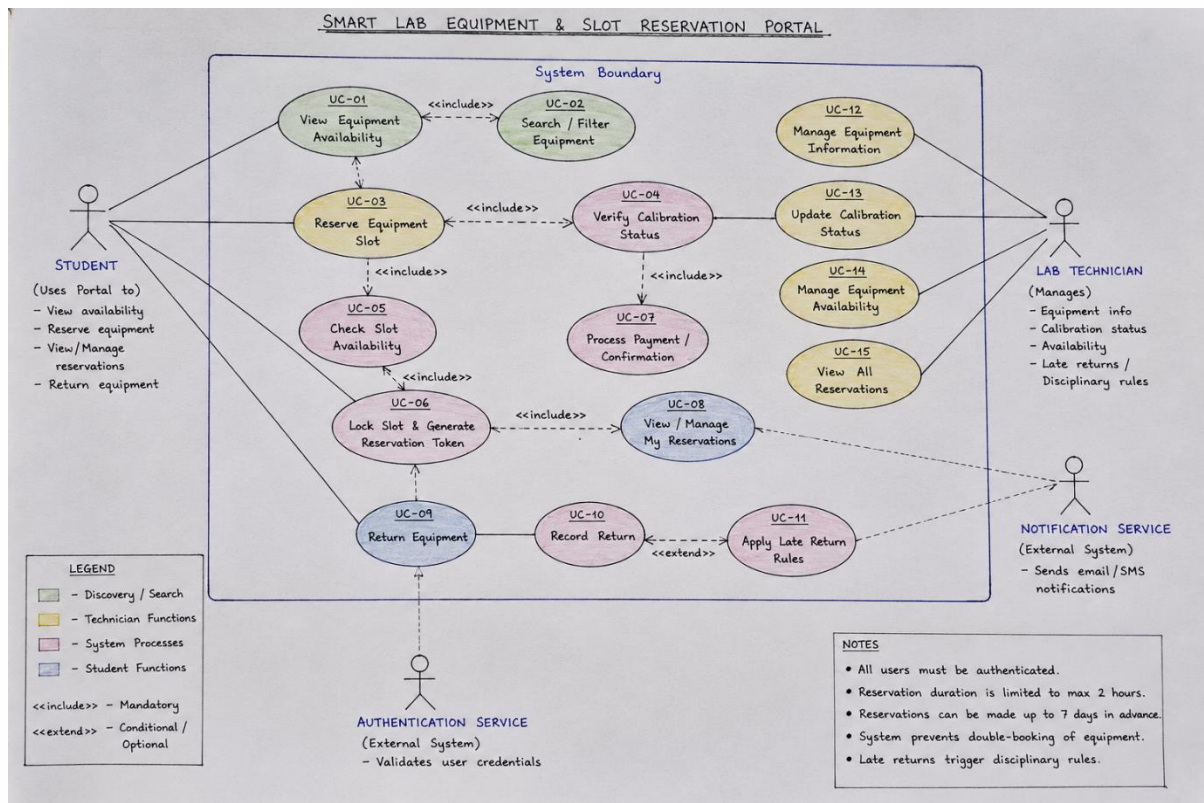
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1. Requirements Table

Five Functional Requirements (FR-001–FR-005, with FR-001 as given) and two Nonfunctional Requirements (NFR-001–NFR-002, with NFR-001 as given).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow authenticated students to view real-time availability and reserve a maximum 2-hour slot for any calibrated equipment up to 7 days in advance.	High	Pass: The slot is successfully locked and a reservation token is generated. Fail: Double-booking is allowed or the reservation exceeds the 2-hour limit.	Core functionality for reserving high-value lab equipment while preventing scheduling conflicts.
FR-002	Functional	The system shall verify the calibration status of selected equipment before confirming a reservation.	High	Pass: A reservation is confirmed only when the selected equipment has valid calibration status. Fail: The system confirms a reservation for equipment whose calibration is invalid or expired.	Ensures students reserve equipment that is currently calibrated and suitable for use.
FR-003	Functional	The system shall prevent two students from reserving the same equipment for overlapping time slots.	High	Pass: When two requests overlap, only one reservation is confirmed and the other request is rejected or redirected to an available slot. Fail: Both overlapping reservations are confirmed.	Prevents double-booking of high-value laboratory equipment.
FR-004	Functional	The system shall allow students to return reserved equipment and record whether the equipment was returned after the reserved slot.	Medium	Pass: A return is recorded with its return time and a late return is identified when applicable. Fail: The system cannot record the return or fails to identify a late return.	Supports equipment tracking and enforcement of late-return rules.
FR-005	Functional	The system shall allow lab technicians to update equipment availability and calibration status.	High	Pass: An authorized lab technician can update an equipment status and the updated status is reflected in the portal. Fail: An unauthorized user can make the update or the updated status is not reflected.	Keeps equipment availability and calibration information accurate for reservations.
NFR-001	Performance & Security	The system shall ensure that concurrent slot reservation requests are processed within 200 ms with strict database-level lock isolation to prevent race conditions.	High	Pass: Benchmarking under simulated peak load confirms the target latency of 200 ms or less and no conflicting reservations are created. Fail: Requests exceed the target latency or race conditions cause conflicting reservations.	Ensures fast, secure, and consistent reservation processing when multiple students attempt to book simultaneously.
NFR-002	Reliability	The system shall preserve confirmed reservation records and reservation tokens without data loss when a reservation transaction is successfully completed.	High	Pass: After a successful reservation, the reservation record and token remain retrievable after logout, refresh, or system restart. Fail: A successfully confirmed reservation or its token is lost.	Ensures confirmed bookings remain reliable and traceable for students and lab technicians.

2. UML Use-Case Diagram



Include relationships:

"View Equipment Availability" includes "Search / Filter Equipment"; "Reserve Equipment Slot" includes "Verify Calibration Status" and "Check Slot Availability"; "Reserve Equipment Slot" includes "Lock Slot & Generate Reservation Token"; "Verify Calibration Status" includes "Process Payment / Confirmation"; "Lock Slot & Generate Reservation Token" includes "View / Manage My Reservations"; "Return Equipment" includes "Record Return".

Extend relationships:

"Manage Return & Late-Return Rules" extends "Record Return" (conditional, only when the equipment is returned late).

3. Use-Case Flow Specification

Use Case: Reserve Equipment Slot

Primary Actor: Student

Secondary Actor: Lab Technician

Precondition: Student is authenticated and has access to the Smart Lab Equipment & Slot Reservation Portal. At least one equipment item is available for reservation, and the requested booking is within the allowed 7-day advance period.

Postcondition (Success): The selected equipment slot is successfully reserved for the student, the slot is locked against other reservations, and a unique reservation token is generated.

Postcondition (Failure): The reservation is not created, the selected slot remains available to other students, and the student is informed of the reason for failure.

Main Success Scenario

1. Student selects **“View Equipment Availability”** from the portal.
2. System displays the available laboratory equipment and corresponding time slots.
3. Student selects the required equipment and an available time slot.
4. System checks the slot availability (<<include>>) to ensure that no overlapping reservation exists.
5. System verifies the calibration status (<<include>>) of the selected equipment.
6. System checks that the requested reservation does not exceed the **2-hour limit** and is not more than **7 days in advance**.
7. System locks the selected equipment slot.
8. System creates the reservation record.
9. System generates a unique reservation token.
10. System displays the reservation confirmation and token to the student.
11. Use case ends successfully.

Alternate Flow

4a. Selected Equipment Slot Is Already Reserved

- **4a1.** System detects that the selected equipment has an overlapping reservation.
- **4a2.** System rejects the reservation request and informs the student that the selected slot is unavailable.
- **4a3.** System displays other available time slots for the selected equipment.
- **4a4.** Student selects an alternative available slot.

- **4a5.** System repeats the slot availability and calibration checks.
- **4a6.** If the alternative slot is valid, the system locks the slot, creates the reservation, and generates a reservation token.
- **4a7.** Use case resumes at Step 9 of the Main Success Scenario.

This follows the same structure as your reference: **Precondition → Success/Failure Postconditions → Main Success Scenario → Alternate Flow**, while using the requirements from your Smart Lab problem statement.